Predicting Academic Functioning in Youth using a Social Determinants of Health Framework

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Background

- **Social Determinants of Health (SDOH)** are the conditions in which people are born, grow, live, work and age, and people's access to power, money and resources (WHO, 2026)
  - Although SDOH provide a useful framework for understanding social inequities, they can be difficult to measure comprehensively.
- **Bourdieu's theory of capitals of Social Space** offers a framework for operationalizing SDOH by capturing resources that shape individuals' opportunities (Bourdieu, 1986)
  - **Economic**: household income, working status, material hardship
  - **Social**: household structure, household size
  - **Cultural**: caregiver educational attainment, language capital, native language
  - **Symbolic**: child ethnic background, racial/ethnic positioning, ethnic identity
- Common SDoH factors that impact **Academic Functioning**:
  - Socioeconomic status (Rakesh et al., 2025)
  - race/ethnicity (Shramko et al., 2024)
- Many studies isolate single variables instead of broader social position
  - There remains limited integration of detailed SDoH profiles with standardized academic assessments, constraining understanding of how social inequities shape academic outcomes.

Research Question

How do the Social, Economic, Cultural, and Symbolic Capital predict performance in reading, math computation, spelling and Pseudoword Decoding?

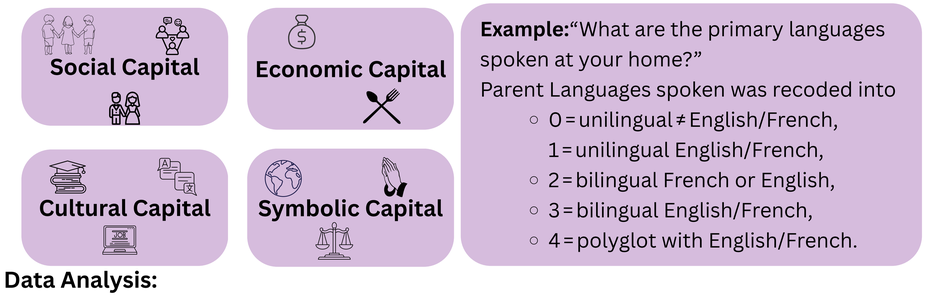
Method

Participants:

- 40 Children aged 8–16 (M age = 12.33, 52% female) and Parent dyads (M age = 35-54, 83% female) from the Greater Montreal Area

Measures:

- Academic achievement: Children's academic achievement was assessed using the Wechsler Individual Achievement Test, Fourth Edition (WIAT-4) measures of Word Reading, Spelling, Numerical Operations, and Pseudoword Decoding (Pearson, 2020).
- SDOH Factors: Parents completed questionnaires including information about social, environmental and family relationships



Example:"What are the primary languages spoken at your home?"
Parent Languages spoken was recoded into

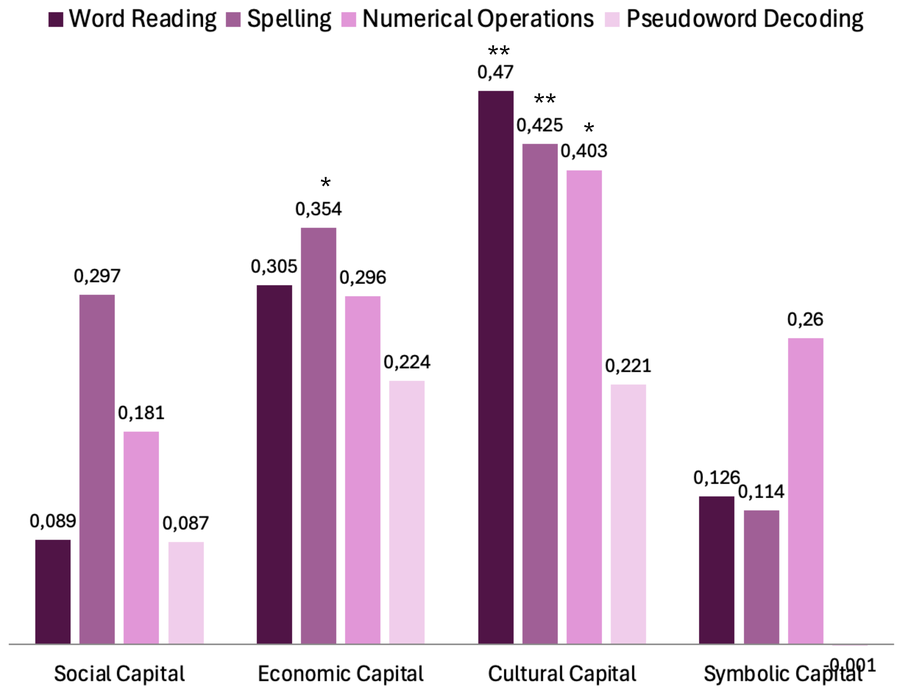
- 0 = unilingual ≠ English/French, 1 = unilingual English/French,
- 2 = bilingual French or English,
- 3 = bilingual English/French,
- 4 = polyglot with English/French.

Data Analysis:

- Recoded socio-demographic factors to create the four index t-scores
- Pearson Correlations and regressions were conducted to understand associations between SDOH indices and WIAT-4 scores of Word Reading, Spelling, Numerical Operations, and Pseudoword Decoding

Results

Figure 1: Correlations between Bourdieu's Four Capitals and WIAT-4 Scores



Note. ** < .01; * < .05. Bars represent Pearson correlation coefficients (r) between Bourdieu's capital T-scores and WIAT-4 scores. Higher values indicate stronger associations.

Overall, the regression model explained 24.1% of the variance in Word Reading R^2 = .241, $F(4,35)$ =2,774, p = .042), and 23.6% of the variance in Spelling (R^2 = .236, $F(4,35)$ =2,703, p = .046), .

- **Social Capital** showed generally weak/moderate associations across WIAT-4 measures
  - Correlations remained non-significant in all domains
  - Did not uniquely predict academic outcomes after controlling for the other capitals (p<.05).
  - → Indicated a limited direct relationship with academic skills.
- **Economic Capital** demonstrated consistently moderate associations with academic outcomes.
  - Relatively stable effects across skills.
  - Its contribution was reduced/non-significant when entered alongside the other capitals (p<.05).
  - → Suggested a broad but less pronounced link to performance.
- **Cultural Capital** consistently showed the strongest positive associations.
  - Correlations ranged from moderate to strong across all Domains
  - After accounting for the overlap among the four forms of capital, Cultural Capital remained the strongest predictor of Word Reading (β = 6.73, p = .03) and Spelling (β = 5.96, p = .03)
- **Symbolic Capital** was the weakest and most inconsistent predictor
  - Showed very small associations overall.
  - Was not a significant independent predictor of academic achievement (p<.05).

Discussion

Key Finding:

- Cultural Capital emerged as the strongest and most consistent predictor of academic achievement.
  - Culturally embedded resources, experiences, and knowledge may play a particularly important role in supporting literacy and numeracy development beyond other forms of capital.
- Economic Capital showed moderate, but less robust, effects, indicating a consistent but weaker contribution to academic achievement.
- Social and Symbolic Capital showed weak or non-significant effects, suggesting limited independent contribution to academic achievement when other forms of capital are included.

Limitations:

- Small sample size limits generalizability of the findings.
- Potential measurement bias in self-reported social capital.

Future Directions:

- Replicate findings in larger cohorts.
- Use longitudinal designs to track developmental trajectories over time.
- Examine additional SDOH factors (e.g., neighborhood safety, education access)

Scan for references

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